

User manual

Every command you type, every button you click, every file that appears, and what each one is for. Written so a person who has never opened a terminal can follow it, and so a judge who asks “show me” gets an answer in under a minute.

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PART 1

Before you begin

You need **two laptops** and **one hotspot**. Nothing else. No internet, no cloud account, no paid software.

NAME	WHICH MACHINE	WHAT IT DOES
Laptop A	the main one, with the project on D:	does the work: wipes, recovers, writes the record
Laptop B	the LOQ	holds a second key and signs. Nothing else lives on it
Hotspot	your phone, or laptop A	lets the two talk. Never campus wi-fi

THE ONE RULE ABOUT CAMPUS WI-FI

Many campus and venue networks block two laptops from seeing each other, even though both show internet. You will not discover this until the moment you need it. Use a

PART 2

Installing, once

Do this the night before, not on the day. Open **PowerShell** on laptop A (press the Windows key, type *powershell*, press Enter).

1 Go to the project folder

```
cd "D:\akhanda SIH\akhanda"
```

Everything in this manual assumes you are here unless it says otherwise.

2 Install what the tool needs

```
pip install -r requirements.txt
```

Takes a minute or two. You only ever do this once.

3 Build the independent checker (optional but worth it)

```
cargo build --release --manifest-path rust/akhanda-verify/Cargo.toml
```

This is a second program that checks the record using completely separate code. It exists so nobody has to take our word for anything.

4 Prove the whole thing works

```
# Windows  
$env:PYTHONPATH="src"; python -m pytest tests/ -q
```

Expect **630 passed** and nothing skipped. This is the number you quote to judges, and it is checked, not claimed.

5 Install the screen (one time)

```
cd "D:\akhanda SIH\frontend"
```

```
npm install
```

Needs Node.js from nodejs.org if you do not have it.

PART 3

Starting the witness laptop

Do this **first**, every time. Everything else depends on it.

1 Join laptop B to the hotspot

Click the wi-fi icon, choose your hotspot, connect.

2 Open the witness folder and start it

```
cd C:\witness_node  
python node.py
```

You must see Application startup complete. **Leave this window open for the entire demo.** Minimise it, never close it.

3 Find laptop B's address

```
ipconfig | findstr /i "IPv4"
```

Write down the number that looks like 192.168.137.132. You need it on laptop A.

MOVE IT OFF THE USB STICK

If you ran it from the USB drive, copy the folder onto laptop B's own hard disk first.
Unplug the stick and the witness dies mid-demo.

PART 4

Running the tool

1 Check laptop A can reach laptop B

```
curl http://192.168.137.132:8000/witness/pubkey
```

You want a short line containing `key_id`. Nothing back means the window in Part 3 is closed, or laptop B's firewall is blocking. See Part 10.

2

Do the work

```
cd "D:\akhanda SIH\akhanda"  
python scripts/demo_bundle.py --witness http://192.168.137.132:8000
```

About five seconds. This one command performs five real operations and writes every file you will show. Watch for all steps ok at the end.

WHAT THAT COMMAND ACTUALLY DID	WHY IT IS IN THE DEMO
Carved four real files out of a disk image	proves recovery returns files byte for byte
Securely erased three files and checked the erase	the erasure half of the problem statement
Tried one job with the witness switched off	it was blocked , and the refusal was recorded
Issued the court certificate	every field marked with the law it answers
Altered a record and re-checked	verification failed and named the record

PART 5

Opening the screen

Two screens exist. Both work. Use whichever you rehearsed.

The main screen

```
cd "D:\akhanda SIH\frontend"  
npm run dev
```

Then open the address it prints, usually `http://localhost:5173`.

The simple screen (no build needed)

```
cd "D:\akhanda SIH\console-app"  
python -m http.server 8080
```

Then open `http://localhost:8080`. This one also lets you open any record file from disk and connect live to laptop B.

KEEP BOTH OPEN IN SEPARATE TABS

They cost nothing to run and either can carry the demo alone.

PART 6

Every screen, button by button

Landing page

The opening screen. Explains in one sentence what the tool does. **Click:** the navigation at the top to move between the other screens.

Console

The record book. Each row is one operation, in order.

WHAT YOU SEE	WHAT IT MEANS
Green dot	this record verifies
Red dot	this record does not verify
WITNESS_COSIGNED	a second key on the other laptop signed it
SOFTWARE_KEY	only this laptop signed it. Honest, and visibly weaker
OPERATION_REFUSED	a job that was blocked and recorded. Point at this one
The long hash	the fingerprint of that record, 64 characters

Live integrity check – the one to demonstrate

BUTTON	WHAT HAPPENS WHEN YOU CLICK IT
Inject Tamper	changes one signed field in memory, then recomputes every fingerprint . Rows turn red because the arithmetic disagrees
Restore Original	puts the original value back and recomputes. Everything returns to green

Each row shows **recorded** and **recomputed** side by side. On a good record they are identical. That is the proof.

IF A JUDGE DOUBTS IT

Press **F12** to open the browser console. Every run prints a table of recorded against recomputed hashes. They are watching the arithmetic happen on their own machine, not reading our claim.

Certificate

The court document. Every field carries the exact law or standard it answers. Scroll to the bottom for the stated limits, which are printed on the certificate itself.

PART 7

Every command, explained

COMMAND	WHAT IT DOES
python node.py	starts the witness on laptop B. Leave open
ipconfig findstr /i "IPv4"	shows a laptop's address on the network
curl http://<ip>:8000/witness/pubkey	asks the witness which key it holds. Your link test
python scripts/demo_bundle.py --witness <url>	runs all five operations and writes every file
python scripts/demo_bundle.py	same, but without a witness. Records the weaker tier honestly
python src/cli.py --no-witness show	prints the record book as text
python src/cli.py --no-witness verify	checks the record book and reports
python -m pytest tests/ -q	runs all 630 tests
python scripts/run_all.py	the full automated check, 11 parts, about six minutes
npm run dev	starts the main screen
python -m http.server 8080	starts the simple screen

TWO THINGS PEOPLE GET WRONG

It is `python src/cli.py`, not `python -m cli`. And on Windows set the path with `$env:PYTHONPATH="src"`, not `export`.

PART 8

Every file, explained

Everything the tool writes lands in `D:\akhanda SIH\akhanda\demo_bundle\`

FILE	WHAT IT IS	SHOWN ON
chain.json	the record book. Every operation, locked to the one before	Console
<code>certificate.json / .txt</code>	the court document, every field marked with its clause	Certificate
<code>byte_identical_proof.json</code>	each recovered file's fingerprint before and after	Evidence
<code>chain_tampered.json</code>	a deliberately altered copy, to prove detection	Integrity
<code>recover_result.json</code>	every file found, and honestly whether any was incomplete	on request
<code>erase_result.json</code>	what was wiped and whether the wipe was checked	on request
<code>MANIFEST.json</code>	which steps ran, which failed. Failures are written down	on request
<code>recovered\</code>	the actual recovered files. You can open them	proof they work

NEVER SHARE THIS ONE FOLDER

`demo_bundle\_akhanda_home\` holds your **private signing key**. Never put it on a USB you hand over, never email it, never upload it. Anyone with that key can sign records as you.

PART 9

Erasing a real device

THIS DESTROYS DATA AND CANNOT BE UNDONE

Only ever do this on a drive you own and have already emptied. Check the drive number twice.

First see which drive is which:

```
Get-Disk
```

Then, and only if you are certain:

```
$env:AKHANDA_ALLOW_DEVICE_WRITE="1"
$env:AKHANDA_TARGET_DEVICE="\\.\PhysicalDrive1"
python src/cli.py erase \\.\PhysicalDrive1 --level Clear `
  --confirm I-UNDERSTAND-THIS-DESTROYS-DATA
```

The tool refuses unless **all three** are correct: the switch, the exact drive path, and the confirm phrase. A typo reaches nothing. Your system disk is refused even when everything is set.

THE HONEST DEMO, WITH NO RISK

Ask for the strongest wipe on a pen drive and the tool records the weaker grade **Clear**, because flash memory keeps blocks no computer can reach. That refusal to overstate is worth more to a judge than any wipe you could show.

PART 10

When something goes wrong

WHAT YOU SEE	WHAT IT MEANS	FIX
Screen says witness unreachable	the window on laptop B was closed	restart python node.py and leave it open
curl hangs or times out	laptop B's firewall is blocking	run node.py again and click Allow , tick Private
Both have internet but cannot see each other	the network blocks laptop-to-laptop	switch to a phone hotspot
Address stopped working	the address changed overnight	ipconfig on laptop B, use the new one

WHAT YOU SEE	WHAT IT MEANS	FIX
No co-signer on any row	you ran without --witness	run again with the witness address
<code>python -m cli</code> not found	wrong command	use <code>python src/cli.py</code>
Screen will not start	dependencies missing	<code>npm install</code> in the frontend folder
Port already in use	it is already running in another window	use the tab you already have open

IF THE HOTSPOT FAILS ENTIRELY

There is a separate plan for that. See **PLAN_B_NO_HOTSPOT** in the same folder. Read it before the day, not during.

PART 11

The five-minute demo script

- 1 **Open the Console. Point at the refused row.**
"That row is an operation that did not happen. The second computer was unreachable, so the tool blocked the job and wrote the refusal down. Ask any other tool to show you the log entry for a job it declined to perform. There isn't one."
- 2 **Show the co-signer on the first row.**
"That key exists only on the other laptop. This machine cannot produce that signature."
- 3 **Go to Live integrity check. Press Inject Tamper.**
"It just recomputed every fingerprint in this browser. One record no longer matches, and everything after it fails too, because each one is locked to the one before. It doesn't say invalid. It says which one."
- 4 **Press F12 if anyone doubts it.**
"Every hash is printed as it is computed. You are watching arithmetic on your own machine, not reading our claim."
- 5 **Press Restore, then open the Certificate.**
"Every field is marked with the exact law or standard it answers. And at the bottom, the certificate states its own limits. A tool that names what it cannot prove is the only kind

worth putting in front of a court.”

Akhanda · अखण्ड, “unbroken” · Syntax Squad · GITAM (Deemed to be University), Hyderabad · Smart India Hackathon 2026, problem statement SIH26149, sponsored by NTRO.

Every command in this manual was run on the machines described before it was written down.